higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 2

(10501042)

25 February 2019 (X-Paper)
09:00–12:00

Candidates need a nonprogrammable scientific calculator and a protractor.

This question paper consists of 8 pages, 1 answer sheet and a formula sheet of 2 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc., which you have used to determine the answers.
5. Answers should be rounded to THREE decimal places, unless stated otherwise.
6. Diagrams are not drawn to scale.
7. Write neatly and legibly.

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QUESTION 1

1.1 Define the following:

1.1.1 Median

1.1.2 Frequency

(2 × 2) (4)

1.2 The heights of a group of boys (to the nearest cm) are given as follows:

139;	148;	159;	149;	162;	157;	158;	164;	173;	181;	166;	168;	168;	168;	189;	174;	188;	177;	185;	179;	187;	179;	183;	168

1.2.1 Calculate the mean heights of the boys. (3)

1.2.2 Arrange the heights in ascending order. (2)

1.2.3 Find the median height of the boys. (3)

1.2.4 Write down the mode height of the boys. (1)

1.3 The following data represents the number of cell phone calls made by 20 teenagers during a month:

27;	54;	34;	67;	48;	63;	43;	49;	56;	35;	51;	47;	45;	58;	41;	46;	52;	67;	53;	30

1.3.1 Calculate the range. (2)

1.3.2 Calculate the value of the lower quartile. (2)

1.3.3 Calculate the value of the upper quartile. (2)

1.3.4 Determine the interquartile range. (2)

1.3.5 Determine the 40th percentile. (3)

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1.4 **QUESTION 1.4.1 and 1.4.2** must be answered on the ANSWER SHEET (attached). Detach it from the question paper, write your EXAMINATION NUMBER in the space provided and hand it in together with your ANSWER BOOK.

A car dealer decided to find out what the most popular colour of car was that customers preferred. He kept a log during a 30 day period of the colours of the cars sold

COLOUR OF CAR SOLD IN A 30 DAY PERIOD				
Silver	Blue	Black	Beige	White
White	Red	White	White	Red
Red	White	Black	Silver	White
Beige	Silver	White	White	Silver
White	Silver	White	Blue	Black
White	White	Blue	Black	Silver

1.4.1 Draw a frequency table **on the answer sheet** using the information given . (7)

1.4.2 Draw a frequency polygon **on the answer sheet** on the grid provided. (7)

1.4.3 What colour did customers seem to prefer? (1)

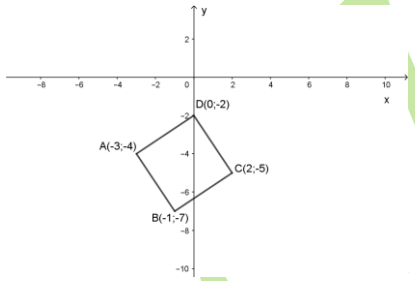
1.4.4 What colour car was sold the least ? (1)

[40]

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QUESTION 2

2.1 Given points A (-3;-4), B(-1;-7) , C(2;-5) and D(0;-2) in the Cartesian plane.



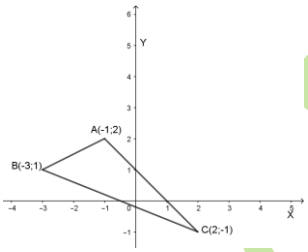
2.1.1 Calculate the perimeter of ABCD. (5)

2.1.2 Calculate M the midpoint of BD. (2)

2.1.3 Calculate the gradient of DC. (2)

2.1.4 Calculate the midpoint of AC.
From this what is M to AC? Give a reason for your answer. (3)

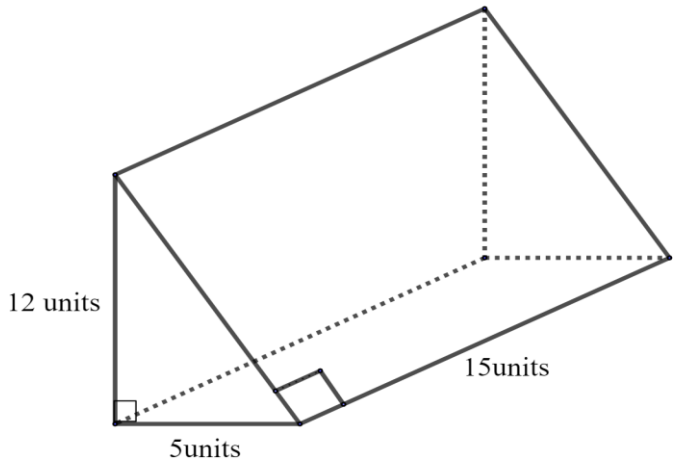
2.2 Given ΔABC in the Cartesian plane with A(-1;2), B(-3;1), C(2;-1).



2.2.1 Describe the translation from A to B. (2)

2.2.2 If ΔABC is reflected about the line $y = x$, what would be the coordinates of the new image $\Delta A^{|}B^{|}C^{|}$? (6)

2.3 A triangular prism is given below. The base of this prism is a right-angled triangle with base 5 units and perpendicular height 12 units. The sides are all rectangles.



2.3.1 Determine the surface area of the triangular prism if all sides are closed. (7)

2.3.2 Determine the volume of this triangular prism. (3)

[30]

QUESTION 3

3.1 $12\tan\theta - 5 = 0$ and $0^0 \le \theta \le 90^0$.

3.1.1 Determine the value of $\sin\theta$.

3.1.2 Determine $\tan^2\theta - \cos^2\theta$.

(2 × 3) (6)

3.2 If $x = 60^0$ *and* $y = 30^0$, then determine the following:

3.2.1 *Cos (x – y)*

3.2.2 *Sin* $(x - 15^0)$

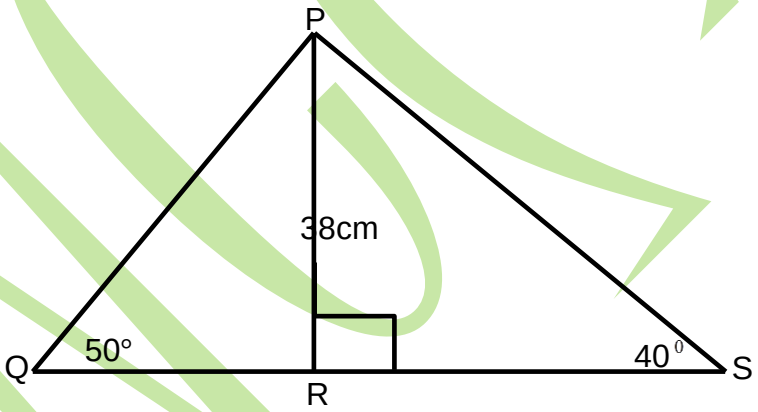
(2 × 2) (4)

3.3 Solve for θ where $\theta \in [0^0; 90^0]$.

$2\cos\theta + 3 = 4$ (3)

3.4 In the diagram PR ⊥ QS, $\angle Q = 50^0$, $\angle S = 40^0$ and PR = 38 cm.

Calculate QS.

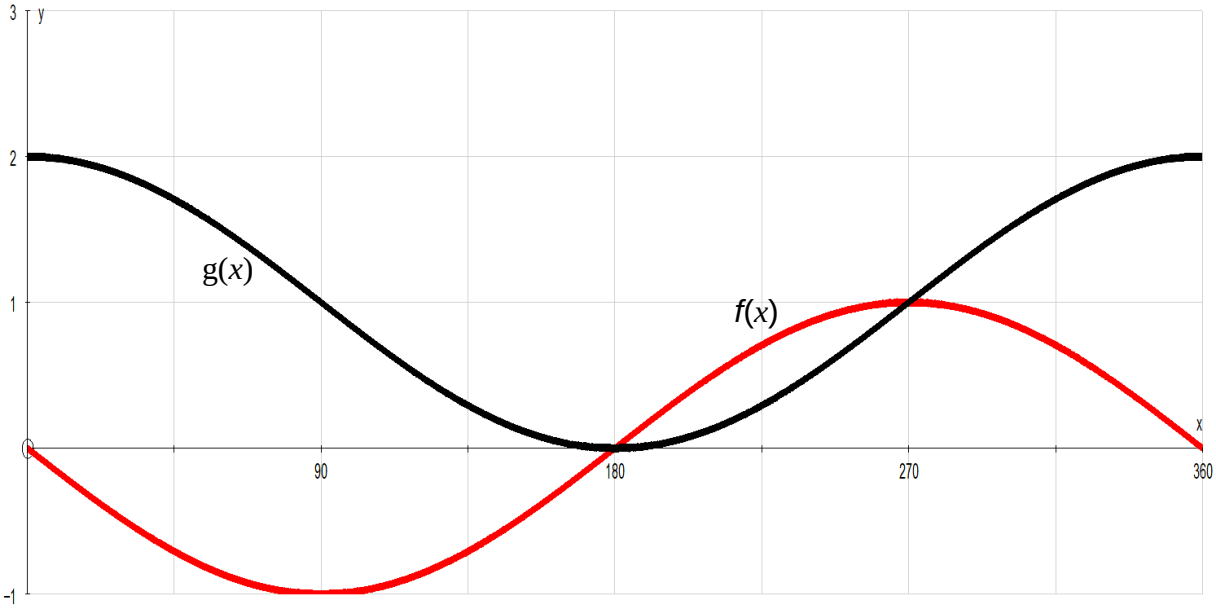


(6)

3.5 The angle of elevation to the top of the building is 82° at a distance of 9 m from the foot of the building.

Calculate the height of the building. (4)

3.6 Below are the graphs of $f(x) = b\sin x$ *and* $g(x) = \cos x + p$.



3.6.1 Determine the equations values of *b* and *p*. (4)

3.6.2 What is the period of g (*x*)? (1)

3.6.3 Write down the range of *f* (*x*) (2)

[30]

TOTAL: 100

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ANSWER SHEET

EXAMINATION NUMBER: | | | | | | | | | | | | | |

QUESTION 1.4

1.4.1

COLOURS OF CARS	TALLYy	FREQUENCY
RED		
BLUE		
SILVER		
WHITE		
BLACK		
BEIGE		
	TOTAL	

(7)

1.4.2

(7)

FORMULA SHEET

1. $A_{square} = l \times l = l^2$

2. $A_{rectangle} = l \times w$

3. $A_{triangle} = \frac{1}{2} b \times h$

4. $A_{circke} = \pi r^2$

5. $C = 2\pi r$

6. Area of a parallelogram = base $\times$ perpendicular height

7. $\mathrm{A}_{hexagon} = \frac{3\sqrt{3}}{2} L^2$

8. $\mathrm{A}_{hexagon} = \frac{\sqrt{3}}{2} W^2$

9. $\mathrm{A}_{cylinder} = 2\pi r(h + r)$

10. Volume = area of base $\times$ perpendicular height

11. Total surface area of a triangular prism = (height of prism $\times$ perimeter of base) + 2(area of base)

12. $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

13. $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

14. $M = (\frac{x_1 + x_2}{2}); \frac{y_1 + y_2}{2})$

15. $\theta = \tan^{-1} m$

16. $\bar{x} = \frac{\sum_{i=1}^{n} x_1}{n}$ or Mean = $\frac{Total\ or\ sum\ of\ all\ items}{number\ of\ items}$

17. $R = X_n - X_1$ or Range = highest value – lowest value

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18. $\bar{x} = \dfrac{\sum f_1 x_1}{n}$

19. $Q_{j \ \ position} = \dfrac{j}{4}(n+1)$

20. Interquartile range = upper quartile – lower quartile = $(Q_3 - Q_1)$

21. Semi-interquartile range = $\dfrac{1}{2}$(upper quartile – lower quartile) = $\dfrac{Q_3 - Q_1}{2}$

22. $P_{j \ \ position} = \dfrac{j}{100}(n+1)$

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